

InterviewIQ Platform

Interview Question Generator, Topic Q&A Knowledge Base, Career Hub & Real-time Grader

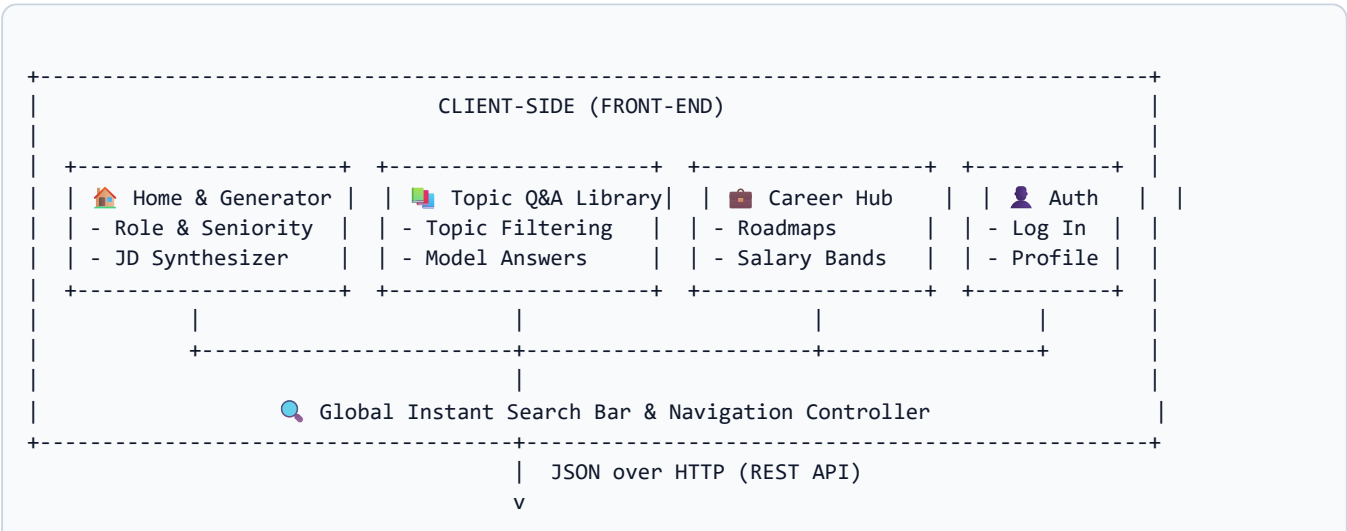
AUTHOR & DEVELOPER	ARCHITECTURE	KEY TOOLS INCLUDED	DATE & STATUS
Yelty Vardhan Reddy	Full-Stack (Node/Express + SPA)	Home, Study Mode, Login, Career, Search, Q&A	September 2026 Production-Ready

1. Project Overview & Core Requirements

InterviewIQ is a full-featured career preparation web platform built for aspiring and junior-to-mid software engineers. It integrates six foundational tools into a single, cohesive user experience:

-  **Back to Home Navigation:** Universal breadcrumbs and return-to-home controllers from every sub-view, ensuring users never feel disoriented while navigating.
-  **Get Ready for Study (Study Mode):** Accelerated 7-day interview sprint schedule, interactive flip flashcards for rapid technical recall, and daily readiness checklist. Accessible via direct deep-link `#get-ready-for-study`.
-  **User Authentication & Profiles (Log In / Sign Up):** Local session persistence, user avatars, 1-click instant demo login, and candidate profile management.
-  **Career Hub & Engineering Roadmaps:** Structured progression ladders for Full Stack, Python, AI/ML, and Web Accessibility engineers, complete with verified salary benchmarks and milestones.
-  **Instant Global Search Engine:** Debounced real-time search across all questions, model answers, tags, and technical concepts.
-  **Topic-Specific Questions & Answers:** Dedicated library allowing candidates to study verified model answers, interviewer intent, and STAR blueprints by specific topic (JavaScript, Python, ML/NLP, SQL & Databases, System Design, Accessibility, Behavioral).

2. System Architecture & High-Level Design



SERVER-SIDE (BACK-END)

Express.js Application Server (Port 3000)
- Static Asset Streaming (HTML/CSS/JS)
- CORS, JSON Body Parsing, Routing Middleware



Question Bank	Question Generator	Answer Evaluator	Auth &
- 8 Core Topics	- JD Tokenizer	- STAR Rubric	Career Hub
- Model Answers	- Heuristic Synthesis	- Metrics & Fluff	Services

3. Detailed Breakdown of the Core Tools

Tool 1: Back to Home Navigation & Breadcrumbs

Navigation consistency is maintained across the entire Single Page Application (SPA). A persistent top-bar brand link, an explicit  [Home](#) navigation button, and a dynamic [← Back to Home](#) breadcrumb bar allow candidates to navigate deep into topic Q&As or career roadmaps and return home with a single click without page reloads.

Tool 2: User Authentication & Profile (Log In / Sign Up)

The authentication subsystem includes:

- **Modal Authentication Dialog:** Supports switching between *Sign In* and *Create Account*.
- **1-Click Instant Demo Login:** Allows users and interviewers to immediately log in as *Yelty Vardhan Reddy* without typing credentials.
- **Session Persistence:** User profile state is stored in `localStorage`, maintaining authenticated state across browser refreshes with avatar badges and a clean *Sign Out* workflow.

Tool 3: Career Hub & Engineering Roadmaps

A dedicated career intelligence center mapping verified market benchmarks:

Role Track	Verified Compensation Range	Core Competencies Required	Key Interview Focus
Full Stack / Web Developer	₹6L – ₹18L PA (India) / \$85k – \$140k (Global)	HTML5, CSS3, JS/TS, Node.js, MySQL, REST, Caching	Event loop, closures, async/await, ACID transactions, and system design.
Python / Backend Engineer	₹6L – ₹22L PA (India) / \$90k – \$150k (Global)	Python 3.x, FastAPI/Django, Concurrency, Docker, SQL	CPython GIL, memory profiling, streaming large datasets, and decorators.
AI / Machine Learning Engineer	₹8L – ₹28L PA (India) / \$110k – \$180k (Global)	Python, Scikit-learn, PyTorch, NLP, Pandas, MLOps	Bias-variance tradeoff, precision/recall, TF-IDF vs embeddings, and latency.
Web Accessibility (a11y) Engineer	₹12L – ₹30L+ PA (Apple, Microsoft ICT2/L4)	WCAG 2.1/2.2, WAI-ARIA, VoiceOver/NVDA, Semantic DOM	First rule of ARIA, focus trapping, modal dialogs, and screen reader trees.

Tool 4: Instant Global Search Engine

A debounced, real-time search bar embedded directly in the header. As the candidate types, the client queries `/api/search?q=...` which searches across question titles, comprehensive model answers, interviewer intents, and skill tags. Results display with matched tags and instant answer reveal accordions.

Tool 5: Topic-Specific Questions & Answers Library

Organized into 8 distinct topic modules (`JavaScript & Web` , `Python Core` , `Machine Learning & NLP` , `Databases & SQL` , `System Design` , `Web Accessibility` , `Behavioral STAR`). Each topic card provides:

1. **The Exact Question:** Real-world scenario or technical concept.

2. 📖 **Comprehensive Model Answer:** Full, verified answer explanation covering mechanics, diagrams, and code snippets.
3. 💡 **Why the Interviewer Asks This:** What underlying skill or red flag the interviewer is grading.
4. 🎯 **Answer Blueprint:** Recommended talking points and structured STAR approach.
5. 💬 **Likely Follow-Up Probes:** 2–3 follow-up questions to prepare for deeper rounds.
6. 🗣️ **Practice & Real-Time Grading:** Live textarea allowing the candidate to test themselves, receiving a score out of 10 with actionable feedback.

Tool 6: Get Ready for Study Mode (7-Day Sprint & Interactive Flashcards)

An accelerated revision and retention center designed for high-stress interview preparation:

- **7-Day Focused Sprint Schedule:** Day-by-day structured curriculum covering Fundamentals, Data Structures & Concurrency, Frameworks & APIs, Database Architecture, System Design & Accessibility, Behavioral STAR Drills, and Mock Interview Dress Rehearsals.
- **Interactive Flip Concept Flashcards:** 3D CSS flip-cards for rapid self-testing on core concepts (e.g., CPython GIL, Event Loop, First Rule of ARIA, Index B-Trees, Precision vs Recall). Candidates click to reveal verified answer summaries.
- **Daily Readiness Checklist:** Interactive checkboxes allowing candidates to track syllabus completion and track daily confidence.
- **Deep-Linking Architecture:** The study view is directly linkable via `#get-ready-for-study`, allowing teachers, mentors, or peers to share the study hub directly.

4. REST API Specification

Method & Endpoint	Purpose & Parameters	Example Response
GET /api/topics	Lists all available topics with question counts.	<code>{"success": true, "topics": [{"id": "python", "questionCount": 3}]}</code>
GET /api/topics/:id	Fetches all questions and model answers for a specific topic.	<code>{"success": true, "topic": {...}, "questions": [...]}</code>
GET /api/search?q=...	Searches across questions, model answers, and tags.	<code>{"success": true, "count": 2, "results": [...]}</code>
GET /api/career/roadmaps	Returns role roadmaps, milestones, and salary benchmarks.	<code>{"success": true, "roadmaps": [...]}</code>
POST /api/auth/login	Authenticates user or creates demo session.	<code>{"success": true, "user": {"name": "Yelty Vardhan Reddy", ...}}</code>
POST /api/generate	Generates tailored questions based on role, level, and JD text.	<code>{"success": true, "meta": {...}, "questions": [...]}</code>
POST /api/evaluate	Grades candidate practice answer (1–10) with rubric feedback.	<code>{"success": true, "evaluation": {"score": 8, "verdict": "..."}}</code>

5. Live Shareable Links & Deployment

Public Shareable URLs (Active Live):

- Direct "Get Ready for Study" Link: <https://0b8a491af6c43422-49-43-233-135.serveusercontent.com/#get-ready-for-study>
- Full Application Main Portal: <https://0b8a491af6c43422-49-43-233-135.serveusercontent.com>
- Local Development Link: <http://localhost:3000/#get-ready-for-study>

How to Run Locally

```
cd interview-question-generator
npm start
# Output:
# =====
# 🚀 Interview Question Generator Server Running!
# 🌐 Local URL: http://localhost:3000
# 📁 Frontend: Serving from ./public
# =====
```